

timer

* Some Concepts

* For pc $\rightarrow$ time unit $\rightarrow$ clk



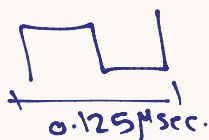
clock $\rightarrow$ some pulse [frequency]

crystal $\rightarrow$ 8 MHz $\rightarrow$ freq = 8 MHz



$$\text{clock time} = \frac{\text{Prescaler}}{\text{system freq}} \rightarrow \frac{1}{8000000}$$

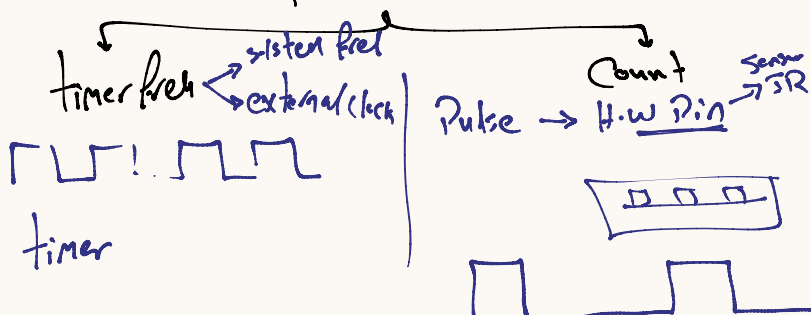
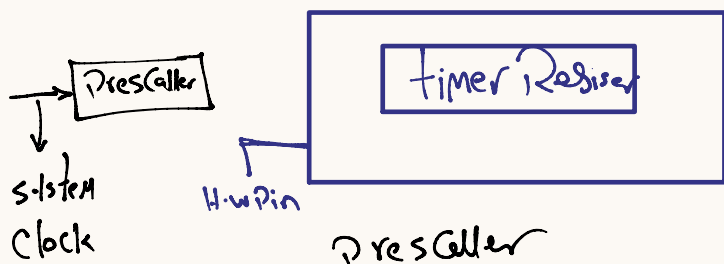
$$= 0.125 \mu\text{sec}$$



led on
delay(1000); $\rightarrow$ Processor is blocked.
led off
delay(1000);

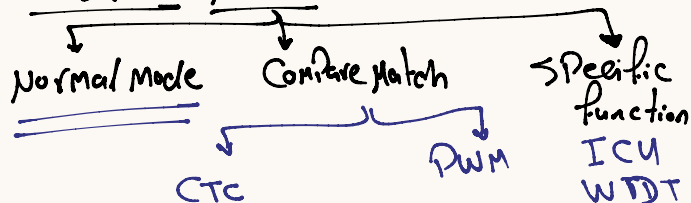
timer/counter

- ① Inputs clocks
- ② Configuration $\rightarrow$ According to alg
- ③ timer $\rightarrow$ flag $\rightarrow$ Interrupt

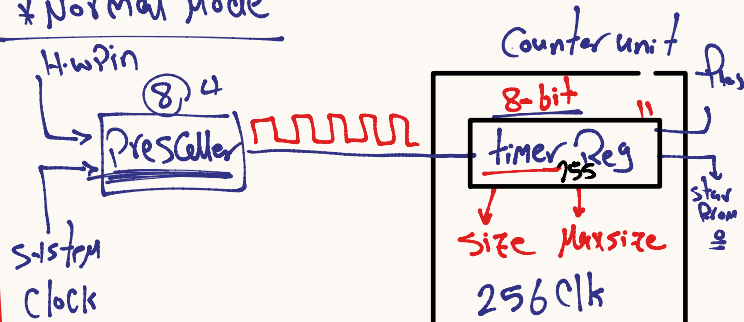


timer/counter

* timer Modes



* Normal Mode



ex: timer 8-bit

$\rightarrow$ No of clk = 256 clk

$\rightarrow$ Max $\Rightarrow$ 255

$\rightarrow$ Min $\Rightarrow$ 0

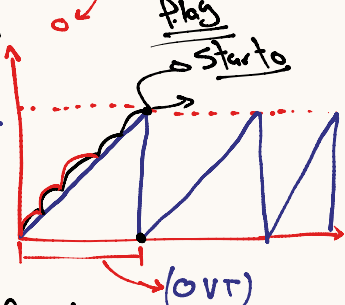
if your Register count to max?

- ① fire overflow flag
- ② start from zero

$\rightarrow$ Max + 1 $\rightarrow$ 255 + 1 $\rightarrow$ char
flag starts

* Normal Mode Calc

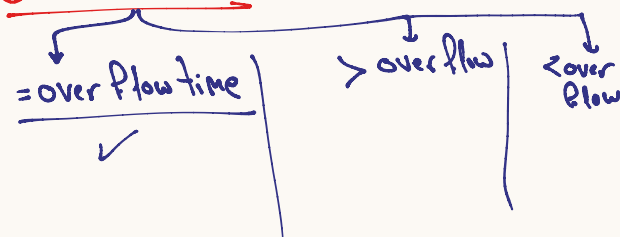
$$\text{clk time} = \frac{\text{tick time Max}}{\text{Prescaler}} \times \text{system freq}$$



② Normal time = overflow time (OVR)

$$= \text{clk time} \times 2^{\text{size}} = \text{clk time} \times 2^8 = \text{clk time} \times 256$$

③ Reg time

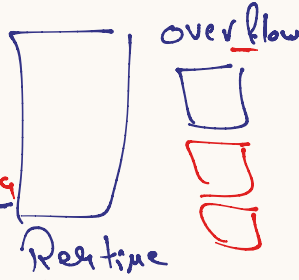


Normal Mode

① $Reg > \text{overflow time}$

No of overflow Count

$$= \frac{Reg \text{ time}}{\text{overflow time}} \approx \frac{1000 \mu\text{sec}}{256 \mu\text{sec}} = 3.9$$



Count $\rightarrow$ Decimal

ISR(overflow)

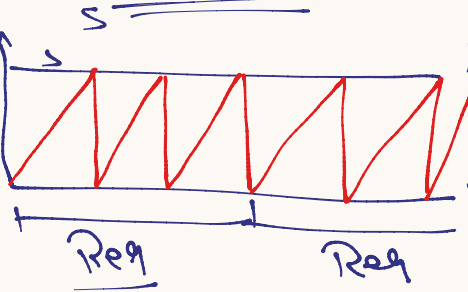
static unsigned int count = 0;

count++;

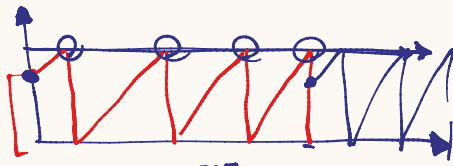
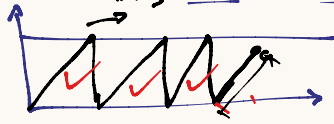
if (count == No of over)

{ Action 1;

count = 0;



Count $\rightarrow$ Floating.



$$\text{Preload} = 2^{\text{size}} * (1 - 0.9) = 2^8 * (1 - 0.9)$$

* the timer Reg shall be updated b-l Preload

timer Init()

{ timer Reg = preload value;

ISR(overflow)

{ static unsigned int count = 0;

count++;

if (count == No of over)

{ Action 1;

timer Reg = preload value;

count = 0;

}

timer 10 bit, Fstop = 8 MHz, Reg = 5 Sec. 100 msec

Prescaler = 8;

① clk time = $\frac{\text{Prescaler}}{F_{stop}} = \frac{8}{8 \text{ MHz}} = 1 \mu\text{sec}$

② overflow time = clk time * 2^{size}
= 1 μsec * 2¹⁰ = 1024 μsec

③ Relation between Reg & overflow

② Reg < overflow

No of Count

$$= \frac{Reg}{\text{overflow}} = \frac{200 \mu\text{sec}}{250 \mu\text{sec}} = 0.8$$



$$* \text{Preload} = 2^{\text{size}} * (1 - 0.9)$$

Init

{ update timer with preload

ISR()

{ static unsigned int count = 0;

if (count == No of over + 1)

④ No of over flow Count

$$= \frac{Reg}{\text{overflow}} = \frac{5 * 10^3}{1024 * 10^{-3}}$$

$$\Rightarrow 4882.8125$$

$$\text{Preload} = 2^{\text{size}} * (1 - 0.9)$$

$$= 2^{10} * (1 - 0.8125)$$

$$= [192]$$

Code

```

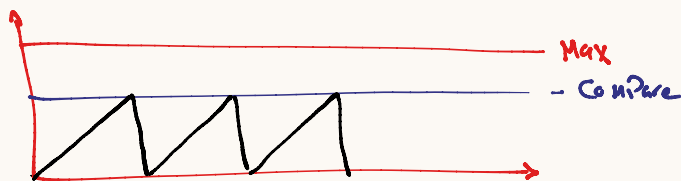
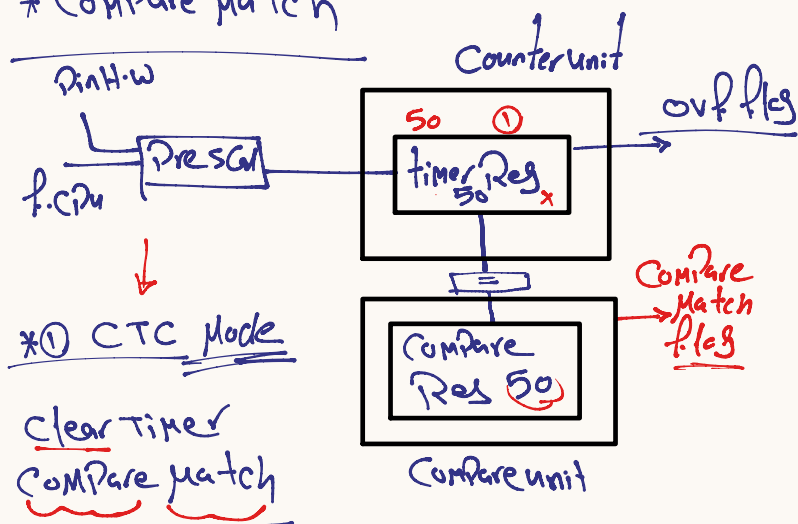
timer_Init()
{
    timer Register = 192;
    ISIR(overflow)
    static ulb counter = 0;
    counter++;
    if (counter == 4883)
    {
        Action();
        timer Reg = 192;
        counter = 0;
    }
}

```

* Normal Mode Calc

- 1) $clk\ time = \frac{Prescaler}{system\ freq.}$
- 2) $overflow\ time = clk\ time \times 2^{size}$
 $= \frac{Prescaler}{system\ freq} \times 2^{size}$
- 3) $\# \text{ No of overflow Count} = \frac{Real\ time}{overflow\ time}$
- 4) $Preload = 2^n \times (1 - 0.5)$

* Compare Match



CTC Mode → When the time Reg match with Compare Reg

- 1) Fire flag (Compare match) flag.
- 2) Clear timer Register.

Compare

No Action → Normal Mode
 Clear → CTC
 toggled → PWM

ex: toggle led each 1000 μsec

timer = 8 f.cpu = 8 MHz Prescaler = 8

1) $tick\ time = \frac{8}{8\ MHz} = 1\ \mu sec.$

2) $compare\ time = tick\ time \times \text{Compare Value}$
 $compare\ time = 1\ \mu sec \times 250 = 250\ \mu sec$

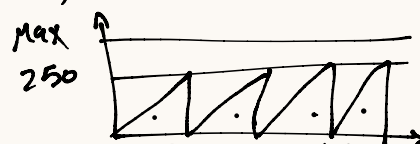
250

Assume compare value

* Rule of Assuming compare

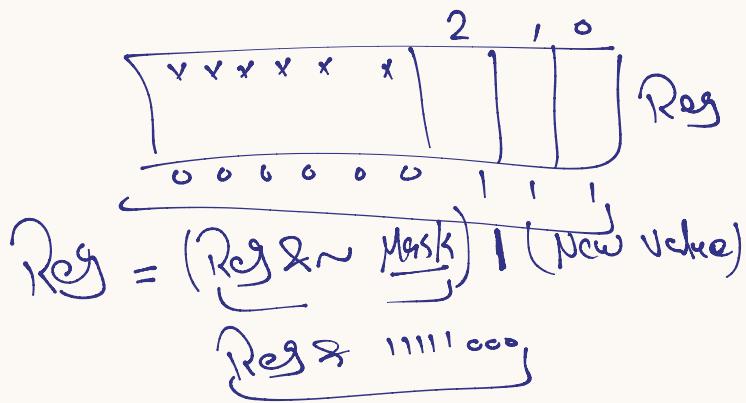
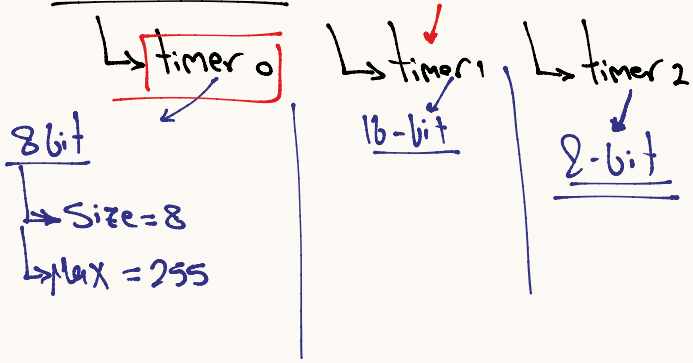
- 1) less than 2^{size}
- 2) No of compare Count is Decided $\frac{Real\ time}{compare\ time}$

$\# \text{ No of Compare} = \frac{Real\ time}{compare\ time}$
 $= \frac{1000\ \mu sec}{250\ \mu sec} = 4$



- 1) $clk\ time = \frac{Prescaler}{f.cpu}$
- 2) $compare\ time = clk\ time \times \text{compare value}$
- 3) $\# \text{ No of compare time} = \frac{Real\ time}{compare\ time}$

Atmega32



Count 4
 3.8 → 3

